Experiment 11: Crypt Arithmetic Problem

Aim:

To solve a crypt arithmetic puzzle (SEND + MORE = MONEY) by assigning a unique digit to every letter so that the arithmetic equation is satisfied.

Algorithm:

1. Collect all distinct letters present in the puzzle words.
2. Generate every possible assignment of distinct digits (0-9) to these letters.
3. Reject an assignment if any leading letter is assigned the digit 0.
4. Convert each word into a number using the assigned digits.
5. Check whether SEND + MORE equals MONEY; if true, return the mapping.
6. If no assignment satisfies the equation, report that no solution exists.

Code:

```
from itertools import permutations

def value(word, m):
    n = 0
    for ch in word:
        n = n * 10 + m[ch]
    return n

def solve(w1, w2, result):
    letters = sorted(set(w1 + w2 + result))
    leading = {w1[0], w2[0], result[0]}
    for perm in permutations(range(10), len(letters)):
        m = dict(zip(letters, perm))
        if any(m[c] == 0 for c in leading):
            continue
        if value(w1, m) + value(w2, m) == value(result, m):
            return m
    return None

w1, w2, res = "SEND", "MORE", "MONEY"
sol = solve(w1, w2, res)
if sol:
    print("Letter mapping:", sol)
    print(w1, "=", value(w1, sol))
    print(w2, "=", value(w2, sol))
    print(res, "=", value(res, sol))
    print(value(w1, sol), "+", value(w2, sol), "=", value(res, sol))
else:
    print("No solution found")
```

Sample Output:

Output

```
Letter mapping: {'D': 7, 'E': 5, 'M': 1, 'N': 6, 'O': 0, 'R': 8, 'S': 9, 'Y': 2}
SEND = 9567
MORE = 1085
MONEY = 10652
9567 + 1085 = 10652
```

Result:

The constraint satisfaction search found the unique valid assignment S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2, giving 9567 + 1085 = 10652. Every letter maps to a different digit and no leading letter is zero.